

一. 单项选择题(共 16 分, 每空满分 2 分)

$$0.4 + 0.3 - 0.12 = 0.58$$

1. 设 A, B 为两个独立的随机事件, $P(A) = 0.4, P(B) = 0.3$, 则 $P(A \cup B) =$

(1); $P(A|B) =$ (2); $P(B|\bar{A}) =$ (3).

C (1) A. 0.6 B. 0.59 C. 0.58 D. 0.57

B (2) A. $\frac{1}{5}$ B. $\frac{2}{5}$ C. $\frac{3}{5}$ D. $\frac{4}{5}$

A (3) A. $\frac{3}{10}$ B. $\frac{4}{10}$ C. $\frac{5}{10}$ D. $\frac{6}{10}$

2. 相互独立的随机变量 X, Y, Z, W , 满足 $X \sim N(5, 10)$ (正态分布),

$Y \sim U(1, 5)$ 均匀分布, Z 服从指数分布且概率密度为 $f_Z(z) = \begin{cases} \beta e^{-\frac{1}{\beta}z}, & z > 0, \\ 0, & \text{其他} \end{cases}$

$$P\{W \leq 1\} = 4e^{-\lambda} = (1 + \lambda)e^{-\lambda}$$

$W \sim \pi(\lambda)$ (泊松分布), $P(W > 1) = 1 - 4e^{-\lambda}$, 则 $\lambda =$ (4); $\beta =$ (5)

$P(Z \leq 18) =$ (6); $E(Z - X - 2Y) =$ (7); $D(-X + Y - eW) =$ (8).

A (4) A. 3 B. 4 C. 5 D. 6

C (5) A. $\frac{1}{3}$ B. $-\frac{1}{3}$ C. $\frac{1}{6}$ D. $-\frac{1}{6}$

A (6) A. $1 - e^{-3}$ B. e^{-3} C. $1 - e^{-6}$ D. e^{-6}

C (7) A. -2 B. -3 C. -5 D. 2

A (8) A. $3e^2 + \frac{34}{3}$ B. $2e^2 + \frac{17}{3}$ C. $3e^2 + \frac{40}{3}$ D. $2e^2 + \frac{40}{3}$

二、(10 分) 二维随机变量 (X, Y) 的联合分布律为

		X		
Y		0	1	2
	0	0	0.2	0.2
	1	0.3	0.1	0.2

求(1) X, Y 的边缘分布律; (2) 写出 $P(X = k | \cdot = 1)$ 的分布律;

(3) 协方差 $\text{Cov}(X, Y)$ 和相关系数 $\rho_{X, Y}$

(4) 判断 X, Y 的相关性和独立性

解: (1) $\begin{array}{c|ccc} X & 0 & 1 & 2 \\ \hline P & 0.3 & 0.3 & 0.4 \end{array}, \begin{array}{c|cc} Y & 0 & 1 \\ \hline P & 0.4 & 0.6 \end{array}$

(2) $\begin{array}{c|ccc} X|Y=1 & 0 & 1 & 2 \\ \hline P & \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \end{array}$

(3)

$$E(X) = 0 \times 0.3 + 1 \times 0.3 + 2 \times 0.4 = 1.1; E(X^2) = 1.9; D(X) = 0.69$$

$$E(Y) = 0 \times 0.4 + 1 \times 0.6 = 0.6; E(Y^2) = 0.6; D(Y) = 0.6 - 0.36 = 0.24$$

$$E(XY) = 1 \times 0.1 + 2 \times 0.2 = 0.5$$

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y) = 0.5 - 1.1 \times 0.6 = -0.16$$

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{D(X)}\sqrt{D(Y)}} = \frac{-0.16}{\sqrt{0.69}\sqrt{0.24}} = \frac{-4\sqrt{46}}{69}$$

(4) 由(3)知 $\rho = \frac{-0.16}{\sqrt{0.69}\sqrt{0.24}} \neq 0$, 所以 X, Y 相关, 不独立。

三、(10分) 设二维随机变量 (X, Y) 的联合分布密度为

$$f(x, y) = \begin{cases} me^{-\frac{y}{3}}, & 0 < x < 3, y > 0 \\ 0, & \text{else} \end{cases}$$

请计算:

1. 常数 m ; 2. 边缘概率密度 $f_X(x)$; 3. 概率 $P(Y \leq X)$; 4.

令 $Z = 2X - Y + 5$, 求 $E(Z), D(Z)$ 。

解: 1. $1 = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = \int_0^3 dx \int_0^{\infty} me^{-\frac{y}{3}} dy = 9m, m = \frac{1}{9}.$

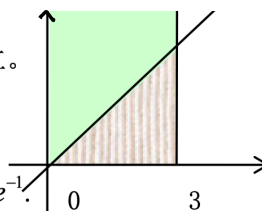
$$2. f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^{\infty} me^{-\frac{y}{3}} dy = 3m = \frac{1}{3}, & 0 < x < 3 \\ 0, & \text{其它} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_0^3 me^{-\frac{y}{3}} dx = 3me^{-\frac{y}{3}} = \frac{1}{3}e^{-\frac{y}{3}}, & y > 0 \\ 0, & \text{其它} \end{cases}$$

$$\therefore f_X(x)f_Y(y) = f(x, y), \text{ 故 } X, Y \text{ 相互独立.}$$

3.

$$P(Y \leq X) = \iint_{y \leq x} f(x, y) dx dy = \int_0^3 dx \int_0^x \frac{1}{9} e^{-\frac{y}{3}} dy = e^{-1}.$$



4. 由 2 问的结论知 $X \sim U(0, 3)$, 故 $E(X) = \frac{3}{2}$, $D(X) = \frac{(3-0)^2}{12} = \frac{3}{4}$;

$Y \sim \text{EXP}(3)$, 故 $E(Y) = 3$, $D(Y) = 3^2 = 9$.

于是 $E(Z) = 2E(X) - E(Y) + 5 = 2 \cdot \frac{3}{2} - 3 + 5 = 5$;

$D(Z) = 2^2 D(X) + (-1)^2 D(Y) + 0 = 4 \cdot \frac{3}{4} + 9 = 12$.

注意: $D(X+Y) = D(X) + D(Y) + 2\text{Cov}(X, Y)$

只有 $\text{Cov}(X, Y) = 0$ 时, 才有 $D(X+Y) = D(X) + D(Y)$.

特别地, X, Y 相互独立时, $\text{Cov}(X, Y) = 0$, 从而 $D(X+Y) = D(X) + D(Y)$.